

Supplementary materials for “On new omnibus tests of uniformity on the hypersphere”

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These supplementary materials contain two sections. Section [A](#) provides the proofs of the results of the paper, which are based on two lemmas. Section [B](#) contains further numerical results omitted from the main text.

Keywords: Directional statistics, Poisson kernel, Sobolev tests, Smooth maximum.

A Proofs

Proof of Proposition 1. Proof of (i). We apply the transformation $t \mapsto (q+1)(\kappa^{-1}t+1)$ to $T_{n,1}$. Since it is monotonic, the tests based on $T_{n,1}$ and based on any statistic resulting from this transformation are equivalent.

First, we address the case $q \geq 2$. Using l'Hôpital's rule and $\partial(z^{-\nu}\mathcal{I}_\nu(z))/\partial z = z^{-\nu}\mathcal{I}_{\nu+1}(z)$,

$$\begin{aligned} \lim_{\kappa \rightarrow 0} (q+1) \left(\frac{e^\kappa}{\kappa} T_{n,1} + 1 \right) \\ = (q+1) \left(1 + \frac{2}{n} \sum_{1 \leq i < j \leq n} \mathbf{X}'_i \mathbf{X}_j - A(n, q) \lim_{\kappa \rightarrow 0} \kappa^{-(q-1)/2} \mathcal{I}_{(q+1)/2}(\kappa) \right), \end{aligned} \quad (15)$$

where $A(n, q) := 2^{(q-3)/2}(n-1)\Gamma((q-1)/2)\Gamma(q-1)$. Using $\mathcal{I}_\nu(z) \asymp (z/2)^\nu/\Gamma(\nu+1)$ when $z \rightarrow 0$ ([DLMF, 2020](#), Equation 10.25.2),

$$\lim_{\kappa \rightarrow 0} \kappa^{-(q-1)/2} \mathcal{I}_{(q+1)/2}(\kappa) = \lim_{\kappa \rightarrow 0} \kappa^{-(q-1)/2} (\kappa/2)^{(q+1)/2} / \Gamma((q+3)/2) = 0,$$

which makes (15) equal to R_n .

Second, we prove the circular case ($q = 1$) analogously, where we use that $\mathcal{I}_0(0) = 1$ and $\mathcal{I}_\nu(0) = 0$, $\nu > 0$. Then,

$$\lim_{\kappa \rightarrow 0} 2 \left(\frac{e^\kappa}{\kappa} T_{n,1} + 1 \right) = 2 \left(1 + \frac{2}{n} \sum_{1 \leq i < j \leq n} \mathbf{X}'_i \mathbf{X}_j - (n-1) \lim_{\kappa \rightarrow 0} \mathcal{I}_1(\kappa) \right) = R_n.$$

Proof of (ii). The proposition holds due to the monotonic relation (5) between $T_{n,1}$ and the LSE function, and the fact that

$$\lim_{\kappa \rightarrow \infty} \text{LSE}_\kappa(\{\mathbf{X}'_i \mathbf{X}_j\}_{1 \leq i < j \leq n}) = \max_{1 \leq i < j \leq n} \mathbf{X}'_i \mathbf{X}_j.$$

□

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Proof of Proposition 2. We apply the transformation $t \mapsto \rho^{-1}(t - 1 + \psi_2(0))$ to $T_{n,2}$. Then,

$$\begin{aligned} \lim_{\rho \rightarrow 0} \frac{T_{n,2} - 1 + \psi_2(0)}{\rho} &= \lim_{\rho \rightarrow 0} \frac{2}{n} \sum_{1 \leq i < j \leq n} \frac{\partial \psi_2(\theta_{ij})}{\partial \rho} + \frac{\partial \psi_2(0)}{\partial \rho} \\ &= (q+1) \left(\frac{2}{n} \sum_{1 \leq i < j \leq n} \mathbf{x}'_i \mathbf{x}_j + 1 \right) = R_n. \end{aligned}$$

Since the transformation is monotonic, the two tests are equivalent. $\square$

Proof of Proposition 3. The coefficients of ψ_1 for $q \geq 2$ follow from the second equation in Magnus et al (1966, page 227):

$$e^{\kappa x} = \left(\frac{2}{\kappa} \right)^{(q-1)/2} \frac{\Gamma((q-1)/2)}{2} \sum_{k=0}^{\infty} (2k+q-1) \mathcal{I}_{(2k+q-1)/2}(\kappa) C_k^{(q-1)/2}(x). \quad (16)$$

Now, taking the limit $q \rightarrow 1$ in (16) produces

$$\begin{aligned} e^{\kappa x} &= \lim_{q \rightarrow 1} \left\{ \frac{(q-1)\Gamma((q-1)/2)}{2} \sum_{k=0}^{\infty} (2k+q-1) \mathcal{I}_{(2k+q-1)/2}(\kappa) (q-1)^{-1} C_k^{(q-1)/2}(x) \right\} \\ &= \sum_{k=0}^{\infty} (2 - \delta_{k,0}) \mathcal{I}_k(\kappa) T_k(x) \end{aligned}$$

after invoking (3) and using $\Gamma(s) = (s + O(s^2))^{-1}$ as $s \rightarrow 0^+$.

The coefficients for ψ_2 follow by combining several generating equations for the Chebyshev and Gegenbauer polynomials. Equation 18.12.7 in DLMF (2020), together with twice Equation 18.12.5 minus Equation 18.12.4, gives

$$\frac{1 - \rho^2}{(1 + \rho^2 - 2\rho x)^{(q+1)/2}} = \begin{cases} \sum_{k=0}^{\infty} (2 - \delta_{k,0}) \rho^k T_k(x), & q = 1, \\ \sum_{k=0}^{\infty} \frac{2k+q-1}{q-1} \rho^k C_k^{(q-1)/2}(x), & q \geq 2. \end{cases}$$

$\square$

Proof of Proposition 4. To derive the expectation of $T_{n,\ell}$, consider the tangent-normal change of variables $\mathbf{x} = x\mathbf{y} + (1-x^2)^{1/2}\mathbf{B}_\mathbf{y}\boldsymbol{\xi}$, with $x := \mathbf{x}'\mathbf{y} \in [-1, 1]$, $\boldsymbol{\xi} \in \mathbb{S}^{q-1}$, and $\mathbf{B}_\mathbf{y}$ a $(q+1) \times q$ matrix with orthonormal columns to $\mathbf{y}$ (see Lemma 2 in García-Portugués et al (2013)). It then follows that

$$\begin{aligned} \mathbb{E}_{\mathcal{H}_0}[\psi_\ell(\theta_{12})] &= \int_{\mathbb{S}^q} \int_{\mathbb{S}^q} \psi_\ell(\cos^{-1}(\mathbf{x}'\mathbf{y})) \, d\nu_q(\mathbf{x}) \, d\nu_q(\mathbf{y}) \\ &= \frac{1}{\omega_q} \int_{\mathbb{S}^q} \int_{\mathbb{S}^{q-1}} \int_{-1}^1 \psi_\ell(\cos^{-1}(x)) (1-x^2)^{q/2-1} \, dx \, d\sigma_{q-1}(\boldsymbol{\xi}) \, d\nu_q(\mathbf{y}) \\ &= \frac{\omega_{q-1}}{\omega_q} \int_{-1}^1 \psi_\ell(\cos^{-1}(x)) (1-x^2)^{q/2-1} \, dx \\ &= b_{0,q}(\psi_\ell). \end{aligned} \quad (17)$$

Plugging this expression into (4) gives $\mathbb{E}_{\mathcal{H}_0}[T_{n,\ell}]$. Recall that the kernel of the U -statistic is centered under $\mathcal{H}_0$: $\tilde{\psi}_\ell = \psi_\ell - b_{0,q}(\psi_\ell)$.

The variance of a U -statistic is (see, e.g., [Serfling, 1980](#), Lemma A, page 183):

$$\mathbb{V}\text{ar}[T_{n,\ell}] = \frac{2(n-1)}{n} [2(n-2)\eta_{1,\ell} + \eta_{2,\ell}], \quad (18)$$

where $\eta_{1,\ell} := \mathbb{V}\text{ar}[\mathbb{E}[\tilde{\psi}_\ell(\theta_{12}) \mid \mathbf{X}_1]]$ and $\eta_{2,\ell} := \mathbb{V}\text{ar}[\tilde{\psi}_\ell(\theta_{12})]$. Under $\mathcal{H}_0$, $\eta_{1,\ell} = 0$ due to the degeneracy of the kernel, and thus

$$\mathbb{V}\text{ar}_{\mathcal{H}_0}[T_{n,\ell}] = \frac{2(n-1)}{n} (b_{0,q}(\psi_\ell^2) - b_{0,q}^2(\psi_\ell)).$$

□

Proof of Theorem 1. The proof follows from [Giné \(1975\)](#) and [Prentice \(1978\)](#). We use the combined key result from these two references, stated in Theorem 3.1 in [García-Portugués et al \(2023\)](#), which establishes that

$$S_{n,q}(\{w_{k,q}\}) \overset{\mathcal{H}_0}{\rightsquigarrow} \sum_{k=1}^{\infty} w_{k,q} Y_{d_{k,q}} \quad (19)$$

for a Sobolev test statistic (1). Due to the connection (7) and (4),

$$S_{n,q}(\{w_{k,q,\ell}\}) = \frac{1}{n} \sum_{i,j=1}^n [\psi_\ell(\theta_{ij}) - b_{0,q}] = \frac{2}{n} \sum_{1 \leq i < j \leq n} \psi_\ell(\theta_{ij}) - nb_{0,q} + \psi_\ell(0).$$

Then, because of Proposition 4 and $\tilde{\psi}_\ell(0) = \psi_\ell(0) - \mathbb{E}_{\mathcal{H}_0}[\psi_\ell(\theta_{12})]$,

$$S_{n,q}(\{w_{k,q,\ell}\}) = T_{n,\ell} + (n-1)\mathbb{E}_{\mathcal{H}_0}[\psi_\ell(\theta_{12})] - nb_{0,q} + \psi_\ell(0) = T_{n,\ell} + \tilde{\psi}_\ell(0)$$

and (9) follows from (19). Note that (9) is also expressible as

$$T_{n,\ell} \overset{\mathcal{H}_0}{\rightsquigarrow} \sum_{k=1}^{\infty} w_{k,q,\ell} (Y_{d_{k,q}} - d_{k,q}),$$

since from (6) evaluated at $\theta = 0$ and expressed in terms of (2), we obtain

$$\sum_{k=1}^{\infty} w_{k,q,\ell} d_{k,q} = \psi_\ell(0) - b_{0,q}(\psi_\ell) = \tilde{\psi}_\ell(0).$$

□

Proof of Corollary 1. Since $b_{k,q}(\psi_\ell) > 0$ for $k \geq 1$, $q \geq 1$, $\kappa > 0$, $0 < \rho < 1$, and $\ell = 1, 2$, the proof follows from [Giné \(1975\)](#) and [Prentice \(1978\)](#); see, e.g., Theorem 3.1 in [García-Portugués et al \(2023\)](#). □

Proof of Proposition 5. For the sake of simplicity, we denote $b_{k,q} \equiv b_{k,q}(\psi_\ell)$. Following the Gegenbauer expansion of g , we can write, for almost every $\mathbf{x} \in \mathbb{S}^q$,

$$f_1(\mathbf{x}) = \begin{cases} \sum_{k=0}^{\infty} e_{k,1} T_k(\mathbf{x}'\boldsymbol{\mu}), & q = 1, \\ \sum_{k=0}^{\infty} e_{k,q} C_k^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}), & q \geq 2. \end{cases} \quad (20)$$

First, we derive the expectation of ψ_ℓ under $\mathcal{H}_1$ for $q \geq 2$:

$$\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12})] = \int_{\mathbb{S}^q} \int_{\mathbb{S}^q} \psi_\ell(\cos^{-1}(\mathbf{x}'\mathbf{y})) f_1(\mathbf{x}) f_1(\mathbf{y}) d\sigma_q(\mathbf{x}) d\sigma_q(\mathbf{y})$$

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$$\begin{aligned}
&= \int_{\mathbb{S}^q} \left[\int_{\mathbb{S}^q} \sum_{k=0}^{\infty} b_{k,q} C_k^{(q-1)/2}(\mathbf{x}'\mathbf{y}) \sum_{m=0}^{\infty} e_{m,q} C_m^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}) d\sigma_q(\mathbf{x}) \right] \\
&\quad \times f_1(\mathbf{y}) d\sigma_q(\mathbf{y}) \\
&= \sum_{k,m=0}^{\infty} b_{k,q} e_{m,q} \int_{\mathbb{S}^q} \omega_q \left[\int_{\mathbb{S}^q} C_k^{(q-1)/2}(\mathbf{x}'\mathbf{y}) C_m^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}) d\nu_q(\mathbf{x}) \right] \\
&\quad \times f_1(\mathbf{y}) d\sigma_q(\mathbf{y}). \tag{21}
\end{aligned}$$

Applying Lemma 1 results in

$$\begin{aligned}
(21) &= \sum_{k=0}^{\infty} b_{k,q} e_{k,q} \omega_q \left(1 + \frac{2k}{q-1} \right)^{-1} \int_{\mathbb{S}^q} C_k^{(q-1)/2}(\mathbf{y}'\boldsymbol{\mu}) f_1(\mathbf{y}) d\sigma_q(\mathbf{y}) \\
&= \sum_{k=0}^{\infty} b_{k,q} e_{k,q}^2 \omega_q^2 \left(1 + \frac{2k}{q-1} \right)^{-1} \int_{\mathbb{S}^q} C_k^{(q-1)/2}(\mathbf{y}'\boldsymbol{\mu}) C_k^{(q-1)/2}(\mathbf{y}'\boldsymbol{\mu}) d\nu_q(\mathbf{y}) \\
&= \sum_{k=0}^{\infty} b_{k,q} e_{k,q}^2 \omega_q^2 \left(1 + \frac{2k}{q-1} \right)^{-2} \frac{(q-1)_k}{k!}, \tag{22}
\end{aligned}$$

where we used that $C_k^{(q-1)/2}(1) = (q-1)_k/k!$ (Table 18.6.1 in [DLMF \(2020\)](#)).

The case $q = 1$ is proved following the same arguments, but using Chebyshev polynomials and the circular version of Lemma 1 twice:

$$\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12})] = \sum_{k=0}^{\infty} \frac{\omega_1^2}{4} (1 + \delta_{k,0})^2 b_{k,1} e_{k,1}^2 T_k(1) = \sum_{k=0}^{\infty} \frac{\omega_1^2}{4} (1 + \delta_{k,0})^2 b_{k,1} e_{k,1}^2. \tag{23}$$

Since f_1 is a pdf, considering a tangent-normal change of variables gives

$$1 = \int_{\mathbb{S}^q} f_1(\mathbf{x}) d\sigma_q(\mathbf{x}) = \omega_{q-1} \int_{-1}^1 g(x)(1-x^2)^{q/2-1} dx = \omega_{q-1} c_{0,1} e_{0,q}$$

and thus $e_{0,q} = 1/\omega_q$ due to (2). Then, to derive the expectation of the statistic, we substitute (17), (22), and (23) into (4), resulting in

$$\begin{aligned}
\mathbb{E}_{\mathcal{H}_1}[T_{n,\ell}] &= (n-1) (\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12})] - \mathbb{E}_{\mathcal{H}_0}[\psi_\ell(\theta_{12})]) \\
&= \begin{cases} \omega_1^2 (n-1)/4 \sum_{k=1}^{\infty} b_{k,1} e_{k,1}^2, & q = 1, \\ \omega_q^2 (n-1) \sum_{k=1}^{\infty} b_{k,q} e_{k,q}^2 \left(1 + \frac{2k}{q-1} \right)^{-2} C_k^{(q-1)/2}(1), & q \geq 2. \end{cases}
\end{aligned}$$

Using the coefficients $\tau_{k,q}$ defined in Proposition 5 concludes the proof. $\square$

Proof of Proposition 6. The variance of a U -statistic is given by (18), where

$$\begin{aligned}
\eta_{1,\ell} &= \mathbb{V}_{\mathcal{H}_1}[\mathbb{E}_{\mathcal{H}_1}[\tilde{\psi}_\ell(\theta_{12}) \mid \mathbf{X}_1]] \\
&= \mathbb{E}_{\mathcal{H}_1}[\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12}) \mid \mathbf{X}_1]^2] - \mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12})]^2 \\
&=: \alpha_{1,\ell} - \beta_{1,\ell}^2
\end{aligned}$$

and similarly $\eta_{2,\ell} = \mathbb{V}_{\mathcal{H}_1}[\psi_\ell(\theta_{12})] =: \alpha_{2,\ell} - \beta_{2,\ell}^2$. The terms $\alpha_{i,\ell}$ and $\beta_{i,\ell}$ can be derived through the Gegenbauer coefficients of ψ_ℓ , denoted by $b_{k,q}$, and the expansion of f_1 in (20).

Applying Lemma 1, we first determine the following conditional expectation:

$$\begin{aligned}\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12}) \mid \mathbf{X}_1] &= \int_{\mathbb{S}^q} \psi_\ell(\mathbf{x}'\mathbf{X}_1) f_1(\mathbf{x}) d\sigma_q(\mathbf{x}) \\ &= \begin{cases} (\omega_1/2) \sum_{k=0}^{\infty} (1 + \delta_{k,0}) b_{k,1} e_{k,1} T_k(\mathbf{X}'_1 \boldsymbol{\mu}), & q = 1, \\ \omega_q \sum_{k=0}^{\infty} b_{k,q} e_{k,q} \left(1 + \frac{2k}{q-1}\right)^{-1} C_k^{(q-1)/2}(\mathbf{X}'_1 \boldsymbol{\mu}), & q \geq 2. \end{cases}\end{aligned}$$

Consider the tangent-normal change of variables $\mathbf{x} = x\boldsymbol{\mu} + (1 - x^2)^{1/2}\mathbf{B}_\mu \boldsymbol{\xi}$ and the expansion (20) for f_1 . Then, for $q \geq 2$,

$$\begin{aligned}\alpha_{1,\ell} &= \mathbb{E}_{\mathcal{H}_1}[(\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12}) \mid \mathbf{X}_1])^2] \\ &= \omega_q^2 \int_{\mathbb{S}^q} \sum_{k_1, k_2=0}^{\infty} b_{k_1,q} b_{k_2,q} e_{k_1,q} e_{k_2,q} \left(1 + \frac{2k_1}{q-1}\right)^{-1} \left(1 + \frac{2k_2}{q-1}\right)^{-1} \\ &\quad \times C_{k_1}^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}) C_{k_2}^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}) f_1(\mathbf{x}) d\sigma_q(\mathbf{x}) \\ &= \omega_q^2 \sum_{k_1, k_2, k_3=0}^{\infty} b_{k_1,q} b_{k_2,q} e_{k_1,q} e_{k_2,q} e_{k_3,q} \left(1 + \frac{2k_1}{q-1}\right)^{-1} \left(1 + \frac{2k_2}{q-1}\right)^{-1} \\ &\quad \times \int_{\mathbb{S}^{q-1}} \int_{-1}^1 C_{k_1}^{(q-1)/2}(x) C_{k_2}^{(q-1)/2}(x) C_{k_3}^{(q-1)/2}(x) (1 - x^2)^{q/2-1} d\sigma_{q-1}(\boldsymbol{\xi}) dx \\ &= \omega_q^2 \omega_{q-1} \sum_{k_1, k_2, k_3=0}^{\infty} b_{k_1,q} b_{k_2,q} e_{k_1,q} e_{k_2,q} e_{k_3,q} \left(1 + \frac{2k_1}{q-1}\right)^{-1} \left(1 + \frac{2k_2}{q-1}\right)^{-1} \\ &\quad \times t_{k_1, k_2, k_3; q},\end{aligned}$$

where $t_{k_1, k_2, k_3; q}$ refers to the triple product of Gegenbauer polynomials defined in Lemma 2 below. Now,

$$\begin{aligned}\beta_{1,\ell} &= \mathbb{E}_{\mathcal{H}_1}[\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12}) \mid \mathbf{X}_1]] \\ &= \omega_q \sum_{k,m=0}^{\infty} b_{k,q} e_{k,q} e_{m,q} \left(1 + \frac{2k}{q-1}\right)^{-1} \int_{\mathbb{S}^q} C_k^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}) C_m^{(q-1)/2}(\mathbf{x}'\boldsymbol{\mu}) d\sigma_q(\mathbf{x}) \\ &= \omega_q^2 \sum_{k=0}^{\infty} b_{k,q} e_{k,q}^2 \left(1 + \frac{2k}{q-1}\right)^{-2} C_k^{(q-1)/2}(1).\end{aligned}$$

For $q = 1$, $\boldsymbol{\xi} \in \mathbb{S}^0 = \{-1, 1\}$ in the tangent-normal change of variables, hence

$$\begin{aligned}\alpha_{1,\ell} &= \mathbb{E}_{\mathcal{H}_1}[(\mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12}) \mid \mathbf{X}_1])^2] \\ &= \frac{\omega_1^2}{2} \sum_{k_1, k_2, k_3=0}^{\infty} (1 + \delta_{k_1,0}) (1 + \delta_{k_2,0}) b_{k_1,1} b_{k_2,1} e_{k_1,1} e_{k_2,1} t_{k_1, k_2, k_3; 1},\end{aligned}$$

where $t_{k_1, k_2, k_3; 1}$ is defined in Lemma 2, and

$$\beta_{1,\ell} = \frac{\omega_1^2}{4} \sum_{k=0}^{\infty} (1 + \delta_{k,0})^2 b_{k,1} e_{k,1}^2.$$

Finally, $\alpha_{2,\ell}$ follows from (22) ($q \geq 2$) and (23) ($q = 1$),

$$\alpha_{2,\ell} = \mathbb{E}_{\mathcal{H}_1}[\psi_\ell^2(\theta_{12})] = \begin{cases} (\omega_1^2/4) \sum_{k=0}^{\infty} (1 + \delta_{k,0})^2 b_{k,1} (\psi_\ell^2) e_{k,1}^2, & q = 1, \\ \omega_q^2 \sum_{k=0}^{\infty} b_{k,q} (\psi_\ell^2) e_{k,q}^2 \left(1 + \frac{2k}{q-1}\right)^{-2} C_k^{(q-1)/2}(1), & q \geq 2, \end{cases}$$

while $\beta_{2,\ell} = \mathbb{E}_{\mathcal{H}_1}[\psi_\ell(\theta_{12})]$ is analogous. $\square$

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Proof of Proposition 7. We have that

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\widehat{\lambda}(\mathcal{S}_1), \mathcal{S}_2) \leq c_\alpha(\widehat{\lambda}(\mathcal{S}_1))] \\
&= \lim_{n \rightarrow \infty} \iint_{\{(\mathbf{x}, \lambda) \in (\mathbb{S}^q)^{|\mathcal{S}_2|} \times \Lambda : T_{|\mathcal{S}_2|, \ell}(\mathbf{x}, \lambda) \leq c_\alpha(\lambda)\}} f_{\mathcal{S}_2}(\mathbf{x}) \, d\mathbf{x} \, dF_{\widehat{\lambda}(\mathcal{S}_1)}(\lambda) \\
&= \lim_{n \rightarrow \infty} \int_{\Lambda} \left[\int_{\{\mathbf{x} \in (\mathbb{S}^q)^{|\mathcal{S}_2|} : T_{|\mathcal{S}_2|, \ell}(\mathbf{x}, \lambda) \leq c_\alpha(\lambda)\}} f_{\mathcal{S}_2}(\mathbf{x}) \, d\mathbf{x} \right] dF_{\widehat{\lambda}(\mathcal{S}_1)}(\lambda) \\
&= \lim_{n \rightarrow \infty} \int_{\Lambda} \mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\lambda) \leq c_\alpha(\lambda)] \, dF_{\widehat{\lambda}(\mathcal{S}_1)}(\lambda),
\end{aligned}$$

where $F_{\widehat{\lambda}(\mathcal{S}_1)}$ is the cdf of the random variable $\widehat{\lambda}(\mathcal{S}_1)$.

Since Λ is a finite set $\{\lambda_1, \dots, \lambda_m\}$ by assumption, interchangeability of the limit and integral is trivial:

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\widehat{\lambda}(\mathcal{S}_1), \mathcal{S}_2) \leq c_\alpha(\widehat{\lambda}(\mathcal{S}_1))] \\
&= \sum_{j=1}^m \lim_{n \rightarrow \infty} \mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\lambda_j) \leq c_\alpha(\lambda_j)] p_{\widehat{\lambda}(\mathcal{S}_1)}(\lambda_j) \\
&= \alpha + \sum_{j=1}^m \lim_{n \rightarrow \infty} \{\mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\lambda_j) \leq c_\alpha(\lambda_j)] - \alpha\} p_{\widehat{\lambda}(\mathcal{S}_1)}(\lambda_j), \quad (24)
\end{aligned}$$

where $p_{\widehat{\lambda}(\mathcal{S}_1)}$ is the probability mass function of $\widehat{\lambda}(\mathcal{S}_1)$ and $\lim_{n \rightarrow \infty} \mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\lambda) \leq c_\alpha(\lambda)] = \alpha$ for all $\lambda \in \Lambda$. From these arguments, $\lim_{n \rightarrow \infty} |\mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\lambda_j) \leq c_\alpha(\lambda_j)] - \alpha| p_{\widehat{\lambda}(\mathcal{S}_1)}(\lambda_j) \leq \lim_{n \rightarrow \infty} |\mathbb{P}_{\mathcal{H}_0}[T_{|\mathcal{S}_2|, \ell}(\lambda_j) \leq c_\alpha(\lambda_j)] - \alpha| = 0$, hence the limit in (24) is zero. $\square$

Remark 3. If Λ is a compact or continuous set, it seems necessary from the proof to control the null asymptotic distribution of $\widehat{\lambda}(\mathcal{S}_1)$. This can be done by an application of the argmax theorem ([van der Vaart and Wellner, 1996](#), Section 3.2.1), which would require establishing the weak convergence of the degenerate U -process $\lambda \mapsto T_{n, \ell}(\lambda)$ (see [Arcones and Giné, 1993](#), Section 5).

Lemma 1 below is a minor extension of Lemma B.7 in [García-Portugués et al \(2023, page 23 of the supplement\)](#) that benefits the paper completeness. It covers the cases $k = 0$ and $m = 0$, as needed in the proof of Propositions 5 and 6.

Lemma 1 (Double products of Gegenbauer and Chebyshev polynomials). *Let $\mathbf{u}, \mathbf{v} \in \mathbb{S}^q$, $q \geq 1$. Then, for $k, m \geq 0$ and $q \geq 2$,*

$$\begin{aligned}
& \int_{\mathbb{S}^1} T_k(\gamma' \mathbf{u}) T_m(\gamma' \mathbf{v}) \, d\nu_1(\gamma) = \frac{1 + \delta_{k,0}}{2} T_k(\mathbf{u}' \mathbf{v}) \delta_{k,m}, \\
& \int_{\mathbb{S}^q} C_k^{(q-1)/2}(\gamma' \mathbf{u}) C_m^{(q-1)/2}(\gamma' \mathbf{v}) \, d\nu_q(\gamma) = \left(1 + \frac{2k}{q-1}\right)^{-1} C_k^{(q-1)/2}(\mathbf{u}' \mathbf{v}) \delta_{k,m}.
\end{aligned}$$

Proof of Lemma 1. The case $k, m \geq 1$ corresponds to Lemma B.7 in [García-Portugués et al \(2023, page 23 of the supplement\)](#). For $k = 0$, since $T_0 \equiv 1$ and

$C_0^{(q-1)/2} \equiv 1$, the orthogonality of the polynomials (when $m \neq 0$) or the integration with respect to the uniform measure ν_q (when $m = 0$) proves the result. $\square$

The final lemma allows the computation of triple products of Gegenbauer and Chebyshev polynomials involved in the proof of Proposition 6.

Lemma 2 (Triple products of Gegenbauer and Chebyshev polynomials). *Let $k_3 \geq k_2 \geq k_1 \geq 0$. Then, for $q \geq 2$,*

$$\begin{aligned} t_{k_1, k_2, k_3; q} &:= \int_{-1}^1 C_{k_1}^{(q-1)/2}(x) C_{k_2}^{(q-1)/2}(x) C_{k_3}^{(q-1)/2}(x) (1-x^2)^{q/2-1} dx \\ &= \begin{cases} 0, & \text{if } s_k \text{ is odd or } k_3 > k_1 + k_2, \\ a_{(k_1+k_2-k_3)/2, k_1, k_2, (q-1)/2} c_{k_3, q}, & \text{if } s_k \text{ is even and } k_3 \leq k_1 + k_2, \end{cases} \end{aligned}$$

where $s_k := \sum_{i=1}^3 k_i$ and

$$a_{\ell, k_1, k_2, \lambda} := \frac{k_1 + k_2 + \lambda - 2\ell}{\ell!(k_1 + k_2 + \lambda - \ell)} \binom{k_1 + k_2 - 2\ell}{k_2 - \ell} \frac{\Gamma(\lambda + \ell) B(\lambda + k_1 - \ell, \lambda + k_2 - \ell)}{\Gamma(\lambda) B(\lambda + k_1 + k_2 - \ell, \lambda)}.$$

In addition,

$$\begin{aligned} t_{k_1, k_2, k_3; 1} &:= \int_{-1}^1 T_{k_1}(x) T_{k_2}(x) T_{k_3}(x) (1-x^2)^{-1/2} dx \\ &= \begin{cases} 0, & \text{if } k_3 \neq k_2 \pm k_1, \\ \frac{1+\delta_{k_1,0}}{2} c_{k_3,1}, & \text{if } k_3 = k_2 \pm k_1. \end{cases} \end{aligned}$$

Proof of Lemma 2. If $k_1 + k_2 < k_3$, the $(k_1 + k_2)$ th order polynomial $x \mapsto C_{k_1}^{(q-1)/2}(x) C_{k_2}^{(q-1)/2}(x)$ can be expressed in terms of $\{C_k^{(q-1)/2}\}_{k=0}^{k_1+k_2}$, and therefore is orthogonal to $C_{k_3}^{(q-1)/2}$. Precisely, the linearization formula for the product of Gegenbauer polynomials (DLMF, 2020, Equation 18.18.22) gives, for $k_2 \geq k_1 \geq 0$ and $\lambda > 0$,

$$C_{k_1}^{(\lambda)}(x) C_{k_2}^{(\lambda)}(x) = \sum_{\ell=0}^{\min(k_1, k_2)} a_{\ell, k_1, k_2, \lambda} C_{k_1+k_2-2\ell}^{(\lambda)}(x),$$

with the positive coefficients

$$\begin{aligned} a_{\ell, k_1, k_2, \lambda} &= \frac{(k_1 + k_2 + \lambda - 2\ell)(k_1 + k_2 - 2\ell)!}{(k_1 + k_2 + \lambda - \ell)\ell!(k_1 - \ell)!(k_2 - \ell)!} \frac{(\lambda)_\ell (\lambda)_{k_1-\ell} (\lambda)_{k_2-\ell} (2\lambda)_{k_1+k_2-\ell}}{(\lambda)_{k_1+k_2-\ell} (2\lambda)_{k_1+k_2-2\ell}} \\ &= \frac{k_1 + k_2 + \lambda - 2\ell}{\ell!(k_1 + k_2 + \lambda - \ell)} \binom{k_1 + k_2 - 2\ell}{k_2 - \ell} \frac{\Gamma(\lambda + \ell) B(\lambda + k_1 - \ell, \lambda + k_2 - \ell)}{\Gamma(\lambda) B(\lambda + k_1 + k_2 - \ell, \lambda)}. \end{aligned}$$

Taking $\lambda = (q-1)/2$ and $k_3 \geq k_2 \geq k_1 \geq 0$ gives

$$C_{k_1}^{(q-1)/2}(x) C_{k_2}^{(q-1)/2}(x) = \sum_{\ell=0}^{k_1} a_{\ell, k_1, k_2, (q-1)/2} C_{k_1+k_2-2\ell}^{(q-1)/2}(x).$$

8 *On new omnibus tests of uniformity on the hypersphere*

Hence, it is clear that the integral is null unless ℓ equals $(k_1 + k_2 - k_3)/2$:

$$\begin{aligned} & \int_{-1}^1 C_{k_1}^{(q-1)/2}(x) C_{k_2}^{(q-1)/2}(x) C_{k_3}^{(q-1)/2}(x) (1-x^2)^{q/2-1} dx \\ &= a_{(k_1+k_2-k_3)/2, k_1, k_2, (q-1)/2} \int_{-1}^1 [C_{k_3}^{(q-1)/2}(x)]^2 (1-x^2)^{q/2-1} dx \\ &= a_{(k_1+k_2-k_3)/2, k_1, k_2, (q-1)/2} c_{k_3, q}. \end{aligned}$$

Analogous derivations are possible for Chebyshev polynomials. From Equation 18.18.21 in [DLMF \(2020\)](#), for $k_2 \geq k_1 \geq 0$,

$$T_{k_1}(x) T_{k_2}(x) = \frac{1}{2} (T_{k_2+k_1}(x) + T_{k_2-k_1}(x))$$

and, consequently,

$$\int_{-1}^1 T_{k_1}(x) T_{k_2}(x) T_{k_3}(x) (1-x^2)^{-1/2} dx = \delta_{k_3, k_2 \pm k_1} \frac{1 + \delta_{k_1, 0}}{2} c_{k_3, 1}.$$

□

B Additional numerical experiments

B.1 Simulation scenarios

To perform the search for the optimal value $\tilde{\lambda}_{\mathcal{H}_1}$ motivated in Section 4, $q_{n, \ell, \mathcal{H}_1}(\lambda)$ must be determined for each alternative distribution. $\text{Var}_{\mathcal{H}_0}[T_{n, \ell}(\lambda)]$ does not depend on the alternative and is given in (8). However, $\mathbb{E}_{\mathcal{H}_1}[T_{n, \ell}(\lambda)]$ does not have a general analytic expression for any $\mathcal{H}_1$. Rotationally symmetric alternatives allow us to directly compute $q_{n, \ell, \mathcal{H}_1}(\lambda)$, since $\mathbb{E}_{\mathcal{H}_1}[T_{n, \ell}(\lambda)]$ is readily available in (10).

We consider rotationally symmetric distributions on $\mathbb{S}^q$ with pdfs

$$\mathbf{x} \mapsto c_{q, \kappa_{\text{dev}}, f} f(\mathbf{x}'\boldsymbol{\theta}; \kappa_{\text{dev}}), \quad (25)$$

where $\boldsymbol{\theta} \in \mathbb{S}^q$ is a location parameter, which is set without loss of generality to $\boldsymbol{\theta} = (0, \dots, 0, 1)'$, $\kappa_{\text{dev}} > 0$ is a concentration parameter, $f : [-1, 1] \times \mathbb{R}_0^+ \rightarrow \mathbb{R}_0^+$ satisfies $f(t; 0) = 1$, and $c_{q, \kappa, f}^{-1} := \int_{-1}^1 (1-t^2)^{q/2-1} f(t; \kappa) dt$ is a normalizing constant. We consider:

- (a) The von Mises–Fisher (vMF) distribution, with $f(t; \kappa) = \exp(\kappa t)$.
- (b) A Cauchy-like (Ca) distribution, with $f(t; \kappa) = (1 - \rho(\kappa)^2) / (1 - 2\rho(\kappa)t + \rho(\kappa)^2)^{(q+1)/2}$, where $\rho(\kappa) := ((2\kappa + 1) - \sqrt{4\kappa + 1}) / (2\kappa)$.
- (c) The Watson (Wa) distribution, with $f(t; \kappa) = \exp(\kappa t^2)$.
- (d) The Small Circle (SC) distribution with $f(t; \kappa, \nu) = \exp(-\kappa(t - \nu)^2)$ and $\nu = 0.25$ being the projection along $\boldsymbol{\theta}$ that controls the modal strip.

In addition, we also consider mixtures consisting of $2(q+1)$ equally-weighted specific distributions with common κ_{dev} and pdfs of the form

$$\mathbf{x} \mapsto \frac{c_{q, \kappa_{\text{dev}}, f}}{2(q+1)} \sum_{k=1}^{2(q+1)} f((-1)^r \mathbf{x}' \mathbf{e}_r; \kappa_{\text{dev}}), \quad r = \lceil k/2 \rceil, \quad (26)$$

with location parameters determined by the r th canonical vector $\mathbf{e}_r$. These mixtures are centro-symmetric about the origin. The distributions used to build the mixtures are:

(e) The vMF distribution (mixture denoted MvMF).

(f) The Cauchy-like distribution (denoted MCa).

For these mixtures $\mathbb{E}_{\mathcal{H}_1}[T_{n,\ell}(\lambda)]$ has no analytic expression available. It is approximated by a Monte Carlo simulation with 10^4 replicates.

Details on how to simulate the above distributions are given in [García-Portugués et al \(2023, Section 4\)](#). The alternative (b) was simulated using the inversion method.

B.2 Precision of approximate oracle parameters

To measure the precision of the oracle parameters in terms of power, we compute the difference between the empirical power attained with $\tilde{\lambda}_{\mathcal{H}_1}$ and the power obtained with the oracle parameter estimated using Monte Carlo, $\lambda_{\mathcal{H}_1}^*$, for the alternatives described in Section B.1. In particular, alternative distributions (a)–(d) with $\kappa_{\text{dev}} \in \{0.5, 1, 1.5, 2\}$ and mixtures (e)–(f) with $\kappa_{\text{dev}} \in \{6, 9, 12, 15\}$ were investigated.

In order to find the oracle parameter $\tilde{\lambda}_{\mathcal{H}_1}$, we consider the grids Λ introduced in Section 4. The only sample size studied was $n = 100$, since n only affects $q_{n,\ell,\mathcal{H}_1}(\lambda)$ through a constant factor, thus keeping $\tilde{\lambda}_{\mathcal{H}_1}$ unchanged for any n . Figures 4 and 5 (left block) show the curves $\lambda \mapsto q_{n,\ell,\mathcal{H}_1}(\lambda)$ suggesting that $\tilde{\lambda}_{\mathcal{H}_1}$ is unique in the alternatives explored.

To estimate the actual oracle parameters, $M = 10^3$ Monte Carlo replicates were drawn from the scenarios described. For each replicate, a test based on $T_{n,\ell}(\lambda)$ at the significance level $\alpha = 0.05$ was performed using the $M = 10^4$ Monte Carlo α -critical values. Then, the empirical power points were smoothed using local constant kernel regression. The right block of Figures 4 and 5 presents the empirical power curve $\lambda \mapsto \mathbb{P}_{\mathcal{H}_1}[T_{n,\ell}(\lambda) > c_{\alpha,n,\ell}(\lambda)]$, which demonstrates that $q_{n,\ell,\mathcal{H}_1}(\lambda)$ captures the behavior of the empirical power of $T_{n,\ell}$ -based tests.

Table 6 shows the difference between the empirical power obtained with $\lambda_{\mathcal{H}_1}^*$ and $\tilde{\lambda}_{\mathcal{H}_1}$. The median power difference for $T_{n,1}$ and $T_{n,2}$ is 0.15% and 0.03%, respectively, which supports the adequacy of the oracle parameter $\tilde{\lambda}_{\mathcal{H}_1}$ obtained with $q_{n,\ell,\mathcal{H}_1}(\lambda)$. Additionally, Table 7 shows the oracle parameters $\tilde{\kappa}_{\mathcal{H}_1}$ and $\tilde{\rho}_{\mathcal{H}_1}$, and the Monte Carlo estimated oracle parameters $\kappa_{\mathcal{H}_1}^*$ and $\rho_{\mathcal{H}_1}^*$ found for alternative distributions (a)–(f) for different concentration values κ_{dev} . The results indicate that $\tilde{\lambda}_{\mathcal{H}_1}$ is not significantly influenced by κ_{dev} .

B.3 Asymptotic null distribution and gamma match accuracies

Tables 8 and 9 contain extended simulation results for the investigation of null tail probabilities computation for $\alpha = 0.01$ and $\alpha = 0.10$, using the asymptotic null distribution computed by Imhof's method, the gamma-match proposed in

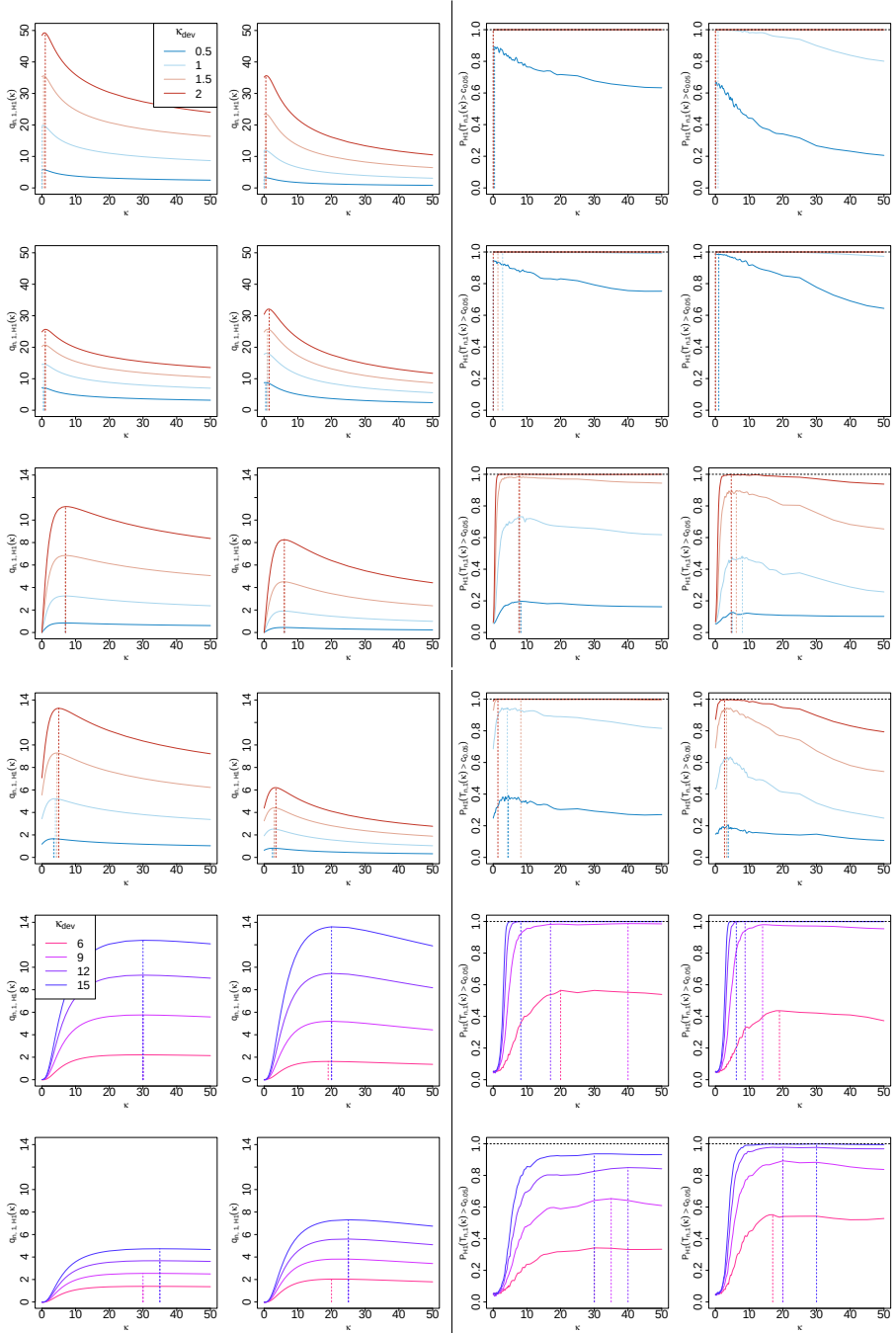


Fig. 4: From top to bottom, rows correspond to rotationally symmetric alternatives (a)–(d) and mixture alternatives (e)–(f). Left/right blocks: curves $\kappa \mapsto q_{n,1,H_1}(\kappa)$ and κ_{H_1} (vertical lines), under alternatives with varying κ_{dev} (acts in (25)–(26)). Right block: empirical smoothed power curves $\kappa \mapsto \mathbb{P}_{H_1}[T_{n,1}(\kappa) > c_{\alpha,n,1}(\kappa)]$ of the $T_{n,1}(\kappa)$ -tests at significance level $\alpha = 0.05$, obtained from $M = 10^3$ samples from the alternative with κ_{dev} , and $\kappa_{H_1}^*$ (vertical lines). The color palettes between the equal-dimension plots in the left and right blocks are shared. The sample sizes are $n = 100$.

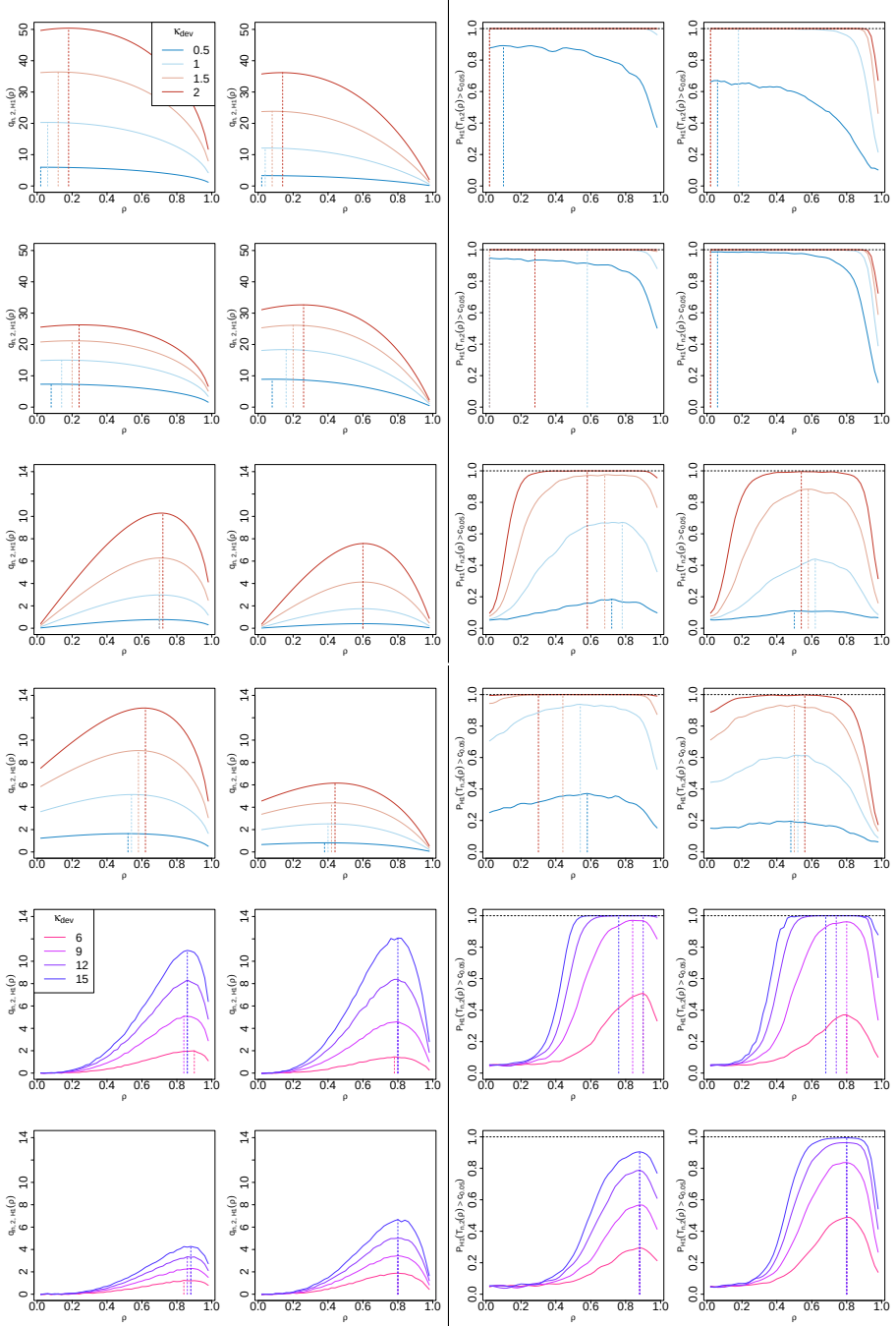


Fig. 5: Analogous description as Figure 4, with $\rho \mapsto q_{n,2,\mathcal{H}_1}(\rho)$ and $\rho \mapsto \mathbb{P}_{\mathcal{H}_1}[T_{n,2}(\rho) > c_{n,2}(\rho)]$.

Alternative	κ_{dev}	$\tilde{\kappa}_{\mathcal{H}_1}$				$\tilde{\rho}_{\mathcal{H}_1}$			
		$q = 1$	$q = 2$	$q = 3$	$q = 5$	$q = 1$	$q = 2$	$q = 3$	$q = 5$
(a) vMF	0.5	1.1	1.3	2.6	0.3	1.4	0.8	0.6	0.1
	1	0.0	0.1	0.3	1.6	0.0	0.1	0.0	1.1
	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
(b) Ca	0.5	0.5	0.2	0.0	0.0	0.6	0.0	0.1	0.0
	1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
(c) Wa	0.5	0.6	0.4	0.0	0.3	0.5	0.5	0.4	0.1
	1	1.6	2.1	1.2	0.8	0.5	0.4	0.0	0.1
	1.5	0.2	1.2	0.8	0.2	0.0	0.1	0.3	0.5
	2	0.0	0.3	0.6	0.0	0.0	0.1	0.8	1.1
(d) SC	0.5	3.5	1.7	1.9	0.7	1.1	0.9	0.4	0.1
	1	0.5	0.6	1.3	1.0	0.0	1.4	1.4	0.0
	1.5	0.0	0.7	2.5	2.5	0.1	0.1	1.1	0.6
	2	0.0	0.2	1.1	0.0	0.0	0.4	0.9	1.5
(e) MvMF	6	0.0	0.1	0.2	0.2	0.0	0.0	0.7	0.2
	9	0.5	0.6	0.6	0.5	0.0	0.0	0.0	0.9
	12	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.1
	15	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
(f) MCa	6	0.0	1.1	0.3	0.4	1.3	0.0	0.0	1.7
	9	1.1	1.3	0.0	0.2	0.0	0.0	0.5	0.2
	12	0.7	0.3	0.1	0.0	0.4	0.0	0.1	0.0
	15	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0

Table 6: Power difference (%) between empirical oracle $\kappa_{\mathcal{H}_1}^*$ and $\rho_{\mathcal{H}_1}^*$ parameters and oracle $\tilde{\kappa}_{\mathcal{H}_1}$ and $\tilde{\rho}_{\mathcal{H}_1}$ parameters for $T_{n,1}$ and $T_{n,2}$, respectively, for alternatives (a)–(f) with κ_{dev} acting in (25)–(26).

Section 6.2, and the Monte Carlo ($M = 10^5$) approximation. The simulation setting is the same as described in Section 6.1.

B.4 Further experiments on the estimation of $\tilde{\lambda}_{\mathcal{H}_1}$

As introduced in Section 5, a first attempt to develop a test based on $T_{n,\ell}$ is to use the two-subsample split approach. The following definition formalizes this procedure.

Definition 3 (Single-partition test of uniformity based on $T_{n,\ell}$). *Given an iid sample $\mathcal{S} = \{\mathbf{X}_1, \dots, \mathbf{X}_n\}$, the $p \cdot 100\%$ -test based on $T_{n,\ell}$ of $\mathcal{H}_0$ at significance level α proceeds as follows:*

1. Split $\mathcal{S}$ into two (disjoint) subsamples $\mathcal{S}_1$ and $\mathcal{S}_2$ of size $|\mathcal{S}_1| = (1 - p)n$ and $|\mathcal{S}_2| = pn$, respectively.
2. Compute $\hat{\lambda}(\mathcal{S}_1) = \arg \max_{\lambda \in \Lambda} T_{|\mathcal{S}_1|,\ell}(\lambda) / \sqrt{\text{Var}_{\mathcal{H}_0}[T_{|\mathcal{S}_1|,\ell}(\lambda)]}$, the estimate of $\tilde{\lambda}_{\mathcal{H}_1}$ using $\mathcal{S}_1$.

Alternative κ_{dev}		$\tilde{\kappa}_{\mathcal{H}_1}(\kappa_{\mathcal{H}_1}^*)$				$\tilde{\rho}_{\mathcal{H}_1}(\rho_{\mathcal{H}_1}^*)$			
		$q = 1$	$q = 2$	$q = 3$	$q = 5$	$q = 1$	$q = 2$	$q = 3$	$q = 5$
(a) vMF	0.5	0.1 (0.4)	0.01 (0.1)	0.01 (0.2)	0.01 (0.1)	0.02 (0.10)	0.02 (0.06)	0.02 (0.08)	0.02 (0.04)
	1	0.2 (0.01)	0.2 (0.8)	0.2 (0.5)	0.1 (0.5)	0.06 (0.02)	0.04 (0.18)	0.02 (0.02)	0.02 (0.12)
	1.5	0.5 (0.01)	0.5 (0.01)	0.2 (0.01)	0.2 (1.1)	0.12 (0.02)	0.08 (0.02)	0.06 (0.02)	0.04 (0.08)
	2	0.7 (0.01)	0.7 (0.01)	0.5 (0.01)	0.5 (0.01)	0.18 (0.02)	0.14 (0.02)	0.08 (0.02)	0.06 (0.02)
(b) Ca	0.5	0.2 (0.01)	0.5 (1)	0.5 (0.3)	0.5 (3.1)	0.08 (0.02)	0.08 (0.06)	0.06 (0.08)	0.08 (0.02)
	1	0.6 (2.8)	0.8 (0.01)	0.9 (0.01)	1.2 (0.01)	0.14 (0.58)	0.16 (0.02)	0.14 (0.02)	0.14 (0.02)
	1.5	1.0 (1.4)	1.0 (0.01)	1.2 (4.1)	1.7 (0.01)	0.20 (0.02)	0.20 (0.02)	0.20 (0.02)	0.20 (0.02)
	2	1.0 (0.01)	1.5 (0.01)	1.7 (0.01)	2.2 (0.01)	0.24 (0.28)	0.26 (0.02)	0.24 (0.02)	0.24 (0.02)
(c) Wa	0.5	6.8 (8.2)	5.6 (4.9)	5.4 (4.9)	5.4 (8.2)	0.70 (0.72)	0.60 (0.50)	0.54 (0.68)	0.46 (0.28)
	1	6.8 (7.6)	5.8 (8.0)	5.4 (7.4)	5.4 (7.2)	0.70 (0.78)	0.60 (0.62)	0.54 (0.56)	0.46 (0.50)
	1.5	7.0 (7.6)	5.8 (6.2)	5.6 (4.8)	5.4 (7.6)	0.70 (0.68)	0.60 (0.58)	0.54 (0.52)	0.46 (0.56)
	2	7.2 (7.8)	6.0 (4.7)	5.6 (4.2)	5.6 (5.6)	0.72 (0.58)	0.60 (0.54)	0.54 (0.58)	0.46 (0.42)
(d) SC	0.5	3.4 (4.5)	2.6 (3.8)	2.3 (2.8)	1.7 (2.8)	0.52 (0.58)	0.38 (0.48)	0.30 (0.40)	0.20 (0.14)
	1	3.8 (4.2)	2.8 (3.4)	2.3 (2.9)	1.7 (2.0)	0.54 (0.54)	0.40 (0.52)	0.30 (0.40)	0.20 (0.20)
	1.5	4.4 (8.2)	3.1 (3.3)	2.4 (3.1)	1.7 (2.1)	0.58 (0.44)	0.42 (0.50)	0.32 (0.34)	0.18 (0.20)
	2	5.0 (1.4)	3.3 (2.7)	2.5 (3.6)	1.7 (1.7)	0.62 (0.30)	0.44 (0.56)	0.32 (0.40)	0.20 (0.24)
(e) MvMF	6	30 (20)	20 (19)	16 (17)	14 (16)	0.90 (0.90)	0.78 (0.80)	0.70 (0.74)	0.64 (0.58)
	9	30 (40)	20 (14)	17 (16)	14 (13)	0.84 (0.84)	0.80 (0.80)	0.72 (0.72)	0.66 (0.64)
	12	30 (17)	20 (9)	17 (18)	15 (12)	0.86 (0.90)	0.80 (0.74)	0.76 (0.66)	0.62 (0.64)
	15	30 (8)	20 (6)	18 (7)	15 (10)	0.86 (0.76)	0.80 (0.68)	0.74 (0.62)	0.68 (0.62)
(f) MCa	6	30 (30)	20 (17)	18 (19)	16 (14)	0.84 (0.88)	0.80 (0.80)	0.72 (0.72)	0.68 (0.64)
	9	30 (35)	25 (20)	20 (20)	18 (14)	0.88 (0.88)	0.80 (0.80)	0.76 (0.72)	0.68 (0.66)
	12	35 (40)	25 (20)	20 (15)	19 (13)	0.86 (0.88)	0.80 (0.80)	0.76 (0.74)	0.68 (0.66)
	15	35 (30)	25 (30)	20 (19)	20 (18)	0.88 (0.88)	0.80 (0.80)	0.76 (0.74)	0.72 (0.62)

Table 7: Oracle $\tilde{\kappa}_{\mathcal{H}_1}$ and $\tilde{\rho}_{\mathcal{H}_1}$ (and Monte Carlo estimated oracle $\kappa_{\mathcal{H}_1}^*$ and $\rho_{\mathcal{H}_1}^*$) parameters for $T_{n,1}$ and $T_{n,2}$, respectively, for alternatives (a)–(f) with κ_{dev} .

3. Use the subsample $\mathcal{S}_2$ to perform the test based on $T_{|\mathcal{S}_2|, \ell}(\hat{\lambda}(\mathcal{S}_1), \mathcal{S}_2)$ at significance level α .

A power investigation of these tests is presented in Figure 6 where the 80%, 75%, and 50%-tests are performed (using gamma-match α -critical values) on 10^3 Monte Carlo replicates of size $n = 100$ for distributions (a)–(f) depending on κ_{dev} . As the curves suggest, the loss of information invested in estimating $\hat{\lambda}_{\mathcal{H}_1}$ reduces the power of the test with respect to the optimal one, especially in multimodal distributions.

References

- Arcones MA, Giné E (1993) Limit theorems for U -processes. *Ann Probab* 21(3):1494–1542. doi:10.1214/aop/1176989128
- DLMF (2020) *NIST Digital Library of Mathematical Functions*. <http://dlmf.nist.gov/>, Release 1.0.27 of 2020-06-15, URL <http://dlmf.nist.gov/>, f. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller and B. V. Saunders, eds.
- García-Portugués E, Crujeiras RM, González-Manteiga W (2013) Kernel density estimation for directional-linear data. *J Multivar Anal* 121:152–175. doi:10.1016/j.jmva.2013.06.009

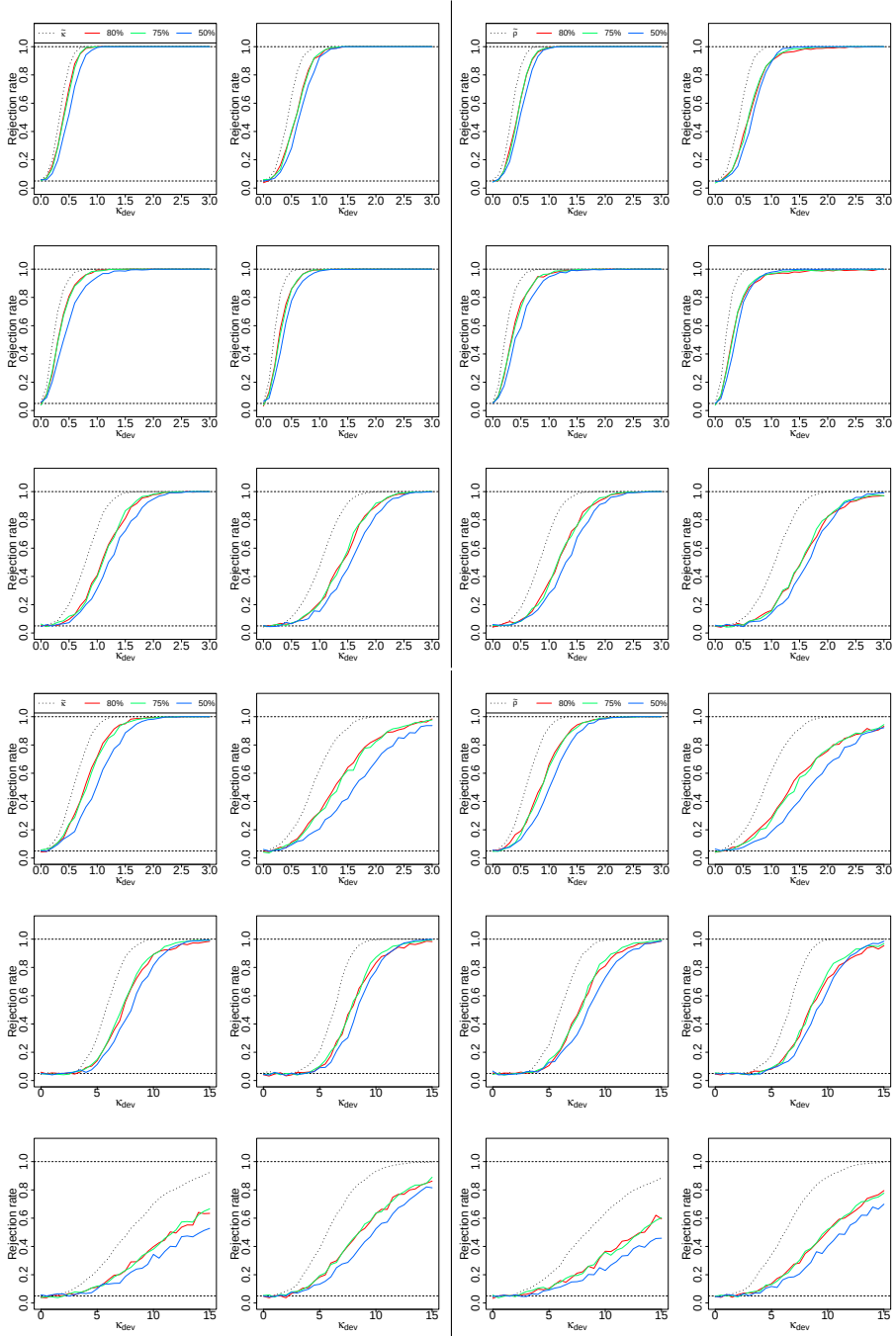


Fig. 6: Empirical rejection frequency of the $p\%$ -test based on $T_{n,1}$ (left panel) and $T_{n,2}$ (right panel) at significance level $\alpha = 0.05$ for alternative distributions (a)–(f) with concentration κ_{dev} . Proportion p is indicated in the legend. $M = 10^3$ samples of size $n = 100$ were drawn from the alternative distribution. Dotted curves indicate the power of the oracle test based on $T_{n,\ell}(\tilde{\lambda}_{\mathcal{H}_1})$.

q	n	Type	$\kappa = 0.1$	$\kappa = 1$	$\kappa = 5$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$
1	10	Asymp.	0.0067	0.0076	0.0087	0.0073	0.0082	0.0097
		Gamma	0.0093	0.0110	0.0124	0.0116	0.0131	0.0149
		MC	0.0104	0.0105	0.0107	0.0105	0.0107	0.0106
	50	Asymp.	0.0099	0.0091	0.0098	0.0096	0.0092	0.0100
		Gamma	0.0101	0.0114	0.0113	0.0119	0.0126	0.0135
		MC	0.0102	0.0101	0.0099	0.0101	0.0104	0.0106
	200	Asymp.	0.0102	0.0096	0.0097	0.0101	0.0099	0.0099
		Gamma	0.0103	0.0119	0.0118	0.0120	0.0122	0.0127
		MC	0.0101	0.0100	0.0097	0.0100	0.0097	0.0097
	10	Asymp.	0.0067	0.0078	0.0101	0.0076	0.0100	0.0154
		Gamma	0.0091	0.0120	0.0154	0.0130	0.0158	0.0234
		MC	0.0101	0.0102	0.0100	0.0103	0.0103	0.0097
2	50	Asymp.	0.0089	0.0094	0.0096	0.0094	0.0099	0.0111
		Gamma	0.0100	0.0126	0.0134	0.0140	0.0146	0.0152
		MC	0.0104	0.0105	0.0108	0.0106	0.0110	0.0106
	200	Asymp.	0.0098	0.0097	0.0099	0.0099	0.0103	0.0102
		Gamma	0.0101	0.0126	0.0125	0.0132	0.0145	0.0138
		MC	0.0107	0.0109	0.0106	0.0111	0.0111	0.0106
	10	Asymp.	0.0063	0.0077	0.0119	0.0079	0.0114	0.0274
		Gamma	0.0088	0.0125	0.0180	0.0143	0.0199	0.0346
		MC	0.0103	0.0104	0.0104	0.0102	0.0106	0.0098
	50	Asymp.	0.0090	0.0092	0.0101	0.0094	0.0103	0.0144
		Gamma	0.0099	0.0133	0.0149	0.0143	0.0158	0.0181
		MC	0.0103	0.0102	0.0099	0.0099	0.0101	0.0106
3	200	Asymp.	0.0098	0.0097	0.0101	0.0099	0.0100	0.0116
		Gamma	0.0106	0.0133	0.0143	0.0150	0.0153	0.0139
		MC	0.0098	0.0099	0.0107	0.0101	0.0102	0.0097
	10	Asymp.	0.0063	0.0075	0.0133	0.0087	0.0173	0.0238
		Gamma	0.0093	0.0128	0.0215	0.0168	0.0274	0.0250
		MC	0.0105	0.0104	0.0104	0.0106	0.0104	0.0095
	50	Asymp.	0.0094	0.0095	0.0105	0.0095	0.0120	0.0304
		Gamma	0.0099	0.0137	0.0157	0.0158	0.0177	0.0315
		MC	0.0100	0.0102	0.0102	0.0104	0.0097	0.0107
	200	Asymp.	0.0097	0.0094	0.0105	0.0102	0.0105	0.0156
		Gamma	0.0101	0.0132	0.0147	0.0155	0.0153	0.0172
		MC	0.0106	0.0106	0.0101	0.0106	0.0106	0.0104

Table 8: Empirical rejection proportion for significance level $\alpha = 0.01$ of $T_{n,1}(\kappa)$ and $T_{n,2}(\rho)$ computed with $M = 10^5$ Monte Carlo samples using each approximation method: asymptotic distribution computed by Imhof's method, our gamma-match approximation, and Monte Carlo ($M = 10^5$) approximation. Boldface denotes that the empirical rejection proportion lies within the 95% confidence interval (0.0092, 0.0108).

García-Portugués E, Navarro-Esteban P, Cuesta-Albertos JA (2023) On a projection-based class of uniformity tests on the hypersphere. *Bernoulli* 29(1):181–204. [doi:10.3150/21-BEJ1454](https://doi.org/10.3150/21-BEJ1454)

Giné E (1975) Invariant tests for uniformity on compact Riemannian manifolds based on Sobolev norms. *Ann Stat* 3(6):1243–1266. [doi:10.1214/aos/1176343283](https://doi.org/10.1214/aos/1176343283)

q	n	Type	$\kappa = 0.1$	$\kappa = 1$	$\kappa = 5$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$
1	10	Asymp.	0.0960	0.0959	0.0903	0.0941	0.0925	0.0893
		Gamma	0.1041	0.1000	0.0957	0.0983	0.0968	0.0979
		MC	0.1007	0.1013	0.1013	0.1017	0.1016	0.1021
	50	Asymp.	0.0995	0.0993	0.0981	0.0991	0.0985	0.0979
		Gamma	0.0998	0.0988	0.0969	0.0955	0.0970	0.0989
		MC	0.1013	0.1007	0.1006	0.1010	0.1010	0.1010
	200	Asymp.	0.1005	0.1006	0.0993	0.1013	0.1006	0.0991
		Gamma	0.0999	0.0978	0.0984	0.0972	0.0968	0.0990
		MC	0.0989	0.1003	0.1015	0.1001	0.1012	0.1015
	10	Asymp.	0.0961	0.0957	0.0891	0.0931	0.0907	0.0909
		Gamma	0.1030	0.1003	0.0969	0.0980	0.0987	0.1022
		MC	0.1006	0.1013	0.1025	0.1016	0.1022	0.0991
2	50	Asymp.	0.0990	0.0990	0.1004	0.0984	0.0978	0.1000
		Gamma	0.0991	0.0972	0.0988	0.0982	0.0990	0.1019
		MC	0.1015	0.1015	0.1022	0.1016	0.1031	0.1027
	200	Asymp.	0.1005	0.0985	0.0995	0.0995	0.0982	0.0993
		Gamma	0.0987	0.1005	0.0995	0.0967	0.1001	0.1006
		MC	0.0999	0.1010	0.1027	0.1005	0.1019	0.1023
	10	Asymp.	0.0943	0.0937	0.0895	0.0914	0.0911	0.0973
		Gamma	0.1020	0.1004	0.0973	0.0989	0.1000	0.1041
		MC	0.1005	0.1006	0.1003	0.1014	0.1001	0.1000
	50	Asymp.	0.0983	0.0996	0.0984	0.0970	0.0984	0.1001
		Gamma	0.0992	0.0996	0.1004	0.1002	0.1007	0.1018
		MC	0.1029	0.1025	0.1039	0.1032	0.1035	0.1017
3	200	Asymp.	0.0997	0.0983	0.1006	0.0994	0.0994	0.1018
		Gamma	0.1000	0.0965	0.0974	0.0983	0.0997	0.1027
		MC	0.0996	0.0995	0.1002	0.1002	0.1004	0.1002
	10	Asymp.	0.0930	0.0917	0.0921	0.0911	0.0943	0.0489
		Gamma	0.1028	0.1014	0.1037	0.1016	0.1039	0.0509
		MC	0.1002	0.1009	0.1011	0.1012	0.1008	0.0994
	50	Asymp.	0.0976	0.0977	0.0982	0.0970	0.0998	0.1004
		Gamma	0.1027	0.1004	0.1043	0.0988	0.1036	0.1023
		MC	0.0985	0.0988	0.1004	0.0989	0.1007	0.0999
	200	Asymp.	0.0995	0.1003	0.0980	0.0995	0.1002	0.1042
		Gamma	0.0997	0.1001	0.1018	0.1008	0.1019	0.1047
		MC	0.1016	0.1020	0.0998	0.1002	0.0983	0.1004

Table 9: Empirical rejection proportion for significance level $\alpha = 0.10$ of $T_{n,1}(\kappa)$ and $T_{n,2}(\rho)$ computed with $M = 10^5$ Monte Carlo samples using each approximation method: asymptotic distribution computed by Imhof's method, our gamma-match approximation, and Monte Carlo ($M = 10^5$) approximation. Boldface denotes that the empirical rejection proportion lies within the 95% confidence interval (0.0984, 0.1015).

Magnus W, Oberhettinger F, Soni RP (1966) Formulas and Theorems for the Special Functions of Mathematical Physics, Die Grundlehren der mathematischen Wissenschaften, vol 52. Springer, Berlin, [doi:10.1007/978-3-662-11761-3](https://doi.org/10.1007/978-3-662-11761-3)

Prentice MJ (1978) On invariant tests of uniformity for directions and orientations. Ann Stat 6(1):169–176. [doi:10.1214/aos/1176344075](https://doi.org/10.1214/aos/1176344075)

- Serfling RJ (1980) Approximation Theorems of Mathematical Statistics. Wiley Series in Probability and Statistics, Wiley, New York, [doi:10.1002/9780470316481](https://doi.org/10.1002/9780470316481)
- van der Vaart AW, Wellner JA (1996) Weak Convergence and Empirical Processes. Springer Series in Statistics, Springer-Verlag, New York, [doi:10.1007/978-1-4757-2545-2](https://doi.org/10.1007/978-1-4757-2545-2)